Zhenjin XU

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EDUCATION

Shanghai University (*University of 211 Project*) Shang Hai, CHINA
Master of Mechatronic Engineering *2018.09-2021.04*
Advisor: Prof. Linyong Shen, Prof. Zhen Zhang; GPA: 3.89/4.0
Course: Programming with MATLAB, C++ primer, Programming with Python, Machine learning

Huaqiao University Xia Men, CHINA
Bachelor of Mechanical Engineering *2014.09-2018.07*
Academic Advisor: Prof. Tingping Lei, Prof. Xiaomei Cai; GPA: 4.3/5.0 (Top 2%)

Xiamen University (*University of 985 Project*) Xia Men, CHINA
PhD. Candidate in Mechatronic Engineering *2021.09-2022.12*
Reason of quitting the PhD: Lab's research does not align with personal career paths.

PUBLICATION

Journal:

[1] **Zhenjin Xu**, Dezhi Wu, Zhiwen Chen, et al. A Flexible Pressure Sensor with Highly Customizable Sensitivity and Linearity via Positive Design of Microhierarchical Structures with a Hyperelastic Model. *Microsystems & Nanoengineering, 2023(9):5* [JCR Q1]

[2] **Zhenjin Xu**, Linyong Shen, Jinwu Qian, et al. Advanced Hand Gesture Prediction Robust to Electrode Shift with an Arbitrary Angle. *Sensors*, 2020(20):4, 1113 [JCR Q2]

[3] Zhongbao Wang, **Zhenjin Xu**, Bin Zhu, et al. Design, fabrication and application of magnetically actuated micro/nanorobots: a review. *Nanotechnology*, 2022(33), 152001. [JCR Q2]

[4] Tingping Lei, **Zhenjin Xu**, Xiaomei Cai, et al. New Insight into Gap Electrospinning: Toward Meter-long Aligned Nanofibers. Langmuir, 2018(34):45, 13788-13793. [JCR Q2]

[5] Tingping Lei, Dahai Lu, **Zhenjin Xu**, et al. 2D -> 3D conversion of superwetting mesh: A simple but powerful strategy for effective and efficient oil/water separation. Separation and Purification Technology. 2019(242), 116244. [JCR Q1]

[6] Yigen Wu, Zhongbao Wang, Guolong Zeng, **Zhenjin Xu**, et al. Printing of Tactile Sensors Upon the Surface of Pneumatic Soft Gripper by Direct Writing and Electrospraying to Enable Intelligent Grasping. Advanced Engineering Materials, 2022, 2200704. [JCR Q2]

[7] Dezhi Wu, Xianshu Cheng, Zhuo Chen, **Zhenjin Xu**, et al. A flexible tactile sensor that uses polyimide/graphene oxide nanofiber as dielectric membrane for vertical and lateral force detection. Nanotechnology, 2022(33):40, 405205. [JCR Q2]

Conference:

[1] **Zhenjin Xu**, Linyong Shen, Wei Song, et al. Unpowered Walking Assistive Device Based on Gait Kinetic Energy Analysis. IEEE 8th Joint International Information Technology and Artificial Intelligence Conference (ITAIC), 2019.

[2] Dezhi Wu, Xianshu Cheng, Yunheng Wu, **Zhenjin Xu***. A Flexible Pressure Sensor with Wide Range Using Polyimide/Graphene Oxide/BaTiO 3 Nanofibers as Dielectric Layer. IEEE 17th International Conference on Nano/Micro Engineered and Molecular Systems (NEMS), 2022. [Corresponding Author]

Patent:

[1] An Unpowered Walking Aid without External Power Input. (ZL201910301835.3)

[2] Determination of Hemispherical Hyperelastic Microstructure for Linear Sensitivity Sensor Design. (ZL202210453163.X)

COMPETITION

- The National Scholarship (2016-2017) (Top 2% of Department)
- Excellent Student in Science and Engineering (2015-2016) (Top 1% of Department)
- First Class Scholarship of the University (2015-2016) (Top 5% of Department)
- First Class Academic Scholarship (2018-2019, 2019-2020, 2020-2021) (Top 5% of Department)

RESEARCH EXPERIENCE

- As **Reviewer** for *Journal of Intelligent & Robotic Systems*, *ACS Applied Materials & Interfaces*, *Sensors and Actuators A*, *Journal of Materials Chemistry A*;

* ***Shanghai University***

Graduate Student (Supervisor: Prof. Linyong Shen)

- Developed an optimization model on real-time hand prediction based on sEMG signals
- Provided a solution to improve the reliability, cost-efficiency and real-time recognition performance in Myo-electric multi-DOFs prosthetic application
- Bio-signal acquisition system development & relevant experiment design and conducting

* ***Huaqiao University***

Research Assistant (Advisor: Prof. Tingping Lei)

- Designed the electrospinning experiment to figure out the roles of solution jet and other governing parameters played
- Utilized gap electrospinning technique we achieved in producing meter-long aligned nanofibers
- Conducted experiments in separating oil/water mixtures using self-made mental mesh coated with specific materials
- Proposed an innovative 3D conversion model and realized in separation the mixtures

* ***Xiamen University***

PhD Candidate (Quit, to 2022.12)

- Proposed a positive design strategy with hyperelastic model for developing sensitivity tuned flexible pressure sensor
- Fabricated prototype sensor verified its ultra-low sensitivity error on targeted requirement over 200 kPa
- Applied the proposed sensor in physiological signal recognition via machine learning
- Lead a research group for developing the high temperature-resistant flexible sensor array for aerospace application
- Conducted experiments on developing ultrahigh temperature-resistance flexible sensor based on EGaIn composites

SKILLS AND INTERESTS

Language: College English Test-6; Chinese (native); IELTS (as soon as possible)

Programming and others: Python, MATLAB, C/ C++ (basic); Origin, Solidworks, AutoCAD.

Interests: Long race (half marathon, PR 2.5h), badminton, coding